Source Sets: A Foundation for Optimal Dynamic Partial Order Reduction

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Stateless model checking is a powerful method for program verification, which however suffers from an exponential growth in the number of explored executions. A successful technique for reducing this number, while still maintaining complete coverage, is Dynamic Partial Order Reduction (DPOR), an algorithm originally introduced by Flanagan and Godefroid in 2005 and since then not only used as a point of reference but also extended by various researchers. In this article, we present a new DPOR algorithm, which is the first to be provably optimal in that it always explores the minimal number of executions. It is based on a novel class of sets, called *source sets*, which replace the role of persistent sets in previous algorithms. We begin by showing how to modify the original DPOR algorithm to work with source sets, resulting in an efficient and simple to implement algorithm, called *source-DPOR*. Subsequently, we enhance this algorithm with a novel mechanism, called *wakeup trees*, that allows the resulting algorithm, called *optimal-DPOR*, to achieve optimality. Both algorithms are then extended to computational models where processes may disable each other, e.g., via locks. Finally, we discuss trade-offs of the source- and optimal-DPOR algorithm and present programs that illustrate significant time and space performance differences between them. We have implemented both algorithms in a publicly available stateless model checking tool for Erlang programs, while the source-DPOR algorithm is at the core of a publicly available stateless model checking tool for C/pthread programs running on machines with relaxed memory models. Experiments show that source sets significantly increase the performance of stateless model checking compared to using the original DPOR algorithm and that wakeup trees incur only a small overhead in both time and space in practice.

CCS Concepts: • **Theory of computation → Verification by model checking**; Logic and verification; • **Software and its engineering →** *Formal software verification*; *Software testing and debugging*;

Additional Key Words and Phrases: dynamic partial order reduction, software model checking, systematic testing, concurrency, source sets, wakeup trees

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